

6.0 HAZARD IDENTIFICATION AND CONTROL PROCEDURE

Purpose

A hazard identification and risk assessment are processes used to identify and evaluate both existing and potential hazards on a worksite and the methods used to control or eliminate the hazards identified.

The assessment process must be completed prior to the start of all jobs to identify existing or potential hazards to workers and eliminate or control these hazards through the use of engineering or administrative controls, proper training or the use of personal protective equipment.

All company staff and contractors are required to take a proactive approach to managing and reporting hazards. When they observe a hazard, they are required to take steps to manage that hazard directly (provided they are adequately knowledgeable / trained to safely do so) – eliminate the hazard or get assistance from appropriate persons to do so whenever reasonably possible. Where hazards cannot be eliminated immediately, take necessary steps to warn others of the hazard. Report hazardous or potentially hazardous conditions and acts to a supervisor or your site contact if a contractor.

Training

All employees will be trained in how to complete a hazard identification and risk assessment at their time of hire and on an annual basis thereafter.

How to complete a Hazard Identification and Risk Assessment

Note: The hazard identification and risk assessment must be completed by a supervisor and communicated to all workers before beginning the work. The JOHSC and workers also play an important role in the assessment process. Being the 'hands-on' people, they often find the new hazards first and then must stop and report. The JOHSC should review Hazard Assessments as part of their routine business at meetings to help make sure that the assessments fit with each other and cover all the company activities.

Step 1: Identify the Hazards

The supervisor is to start the hazard identification process before the job begins by identifying hazards that are known to exist on site and documenting them. By identifying hazards early, the supervisor may be able to implement controls before any workers arrive on site.

To ensure the process is thorough the supervisor should:

- look at all aspects of the work,
- include non-routine activities such as maintenance, repair, or cleaning,

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- look at accident / incident / near-miss records, include people who work "offsite" either at home, on other job sites, drivers, teleworkers, with clients, etc.,
- look at the way the work is organized or "done" (include experience and age of people doing the work, systems being used, etc),
- look at foreseeable unusual conditions (for example: possible impact on hazard control procedures that may be unavailable in an emergency situation, power outage, etc.),
- examine risks to visitors or the public,
- include an assessment of groups that may have a different level of risk such as young or inexperienced workers, persons with disabilities, or new or expectant mothers.

To identify potential hazards, the following questions may be asked (this is not a complete list):

1. Can any body part get caught in or between objects?
2. Do tools, machines or equipment present any hazards?
3. Can the worker make harmful contact with objects?
4. Can the worker slip, trip or fall?
5. Can the worker suffer a strain from lifting, pushing or pulling?
6. Is the worker exposed to extreme heat or cold?
7. Is excessive noise or vibration a problem?
8. Is there a danger from falling objects?
9. Is lighting a problem?
10. Can weather conditions affect safety?
11. Is harmful radiation a possibility?
12. Can contact be made with hot, toxic or caustic substances?
13. Are there dusts, fumes, mists or vapors in the air?

Step 2: Assess the Risk

Ranking or prioritizing hazards is one way to help determine which hazard is the most serious and thus which hazard to control first. Priority is usually established by taking into account the employee exposure and the potential for accident, injury or illness. By assigning a priority to the hazards, you are creating a ranking or an action list. The following factors play an important role:

- percentage of workforce exposed,
- frequency of exposure,
- degree of harm likely to result from the exposure,
- probability of occurrence.

There is no one simple or single way to determine the level of risk. Ranking hazards requires the knowledge of the workplace activities, urgency of situations, and most importantly, objective judgment. One option is seen in the following two tables:

<u>Risk Severity Index</u>		
<u>1</u>	<u>Level 1</u>	<u>Fatality OR Property Damage Exceeding \$50,000</u>
	<u>Level 2</u>	<u>Employee admitted to hospital or probably permanent disability OR property damage between \$10,000 and \$50,000</u>
	<u>Level 3</u>	<u>Employee not able to perform all of their regular duties OR property damage between \$1,000 and \$10,000</u>
	<u>Level 4</u>	<u>Employee able to perform all their regular duties OR property damage less than \$1,000</u>
<u>Probability Index of Occurrence</u>		<u>Example</u>
	<u>A</u>	<u>Likely to occur immediately</u>
		<u>Could happen any day</u>

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2	<u>B</u> <u>Probable in time</u>	<u>Likely to happen if conditions are repeated</u>
	<u>C</u> <u>Possible in time</u>	<u>Under the right conditions, the incident might be repeated</u>
	<u>D</u> <u>Remotely possible</u>	<u>Even under similar conditions, it is unlikely the incident will be repeated</u>

For the activity being examined, determine the most likely reasonable level of severity (levels 1 through 4 in the table on page 10). Then determine how likely (the probability) the injury would be (letters A-D).

		<u>Probability of Occurrence</u>			
		<u>A</u>	<u>B</u>	<u>C</u>	<u>D</u>
<u>Potential Severity</u>	<u>1</u>	<u>1</u>	<u>1</u>	<u>1</u>	<u>2</u>
	<u>2</u>	<u>1</u>	<u>2</u>	<u>2</u>	<u>3</u>
	<u>3</u>	<u>2</u>	<u>2</u>	<u>3</u>	<u>3</u>
	<u>4</u>	<u>2</u>	<u>3</u>	<u>3</u>	<u>4</u>

<u>Risk Definitions</u>	<u>Risk is the chance or probability that a person will be harmed or experience an adverse health effect if exposed to a hazard. It may also apply to situations with property or equipment loss.</u>
<u>4 - Low</u>	<u>Activities in this category contain minimal risk and are unlikely to occur. Organizations can proceed with these activities as planned.</u>
<u>3 - Medium</u>	<u>Activities in this category contain minor to serious risks that are remotely likely to likely to occur. Application of proactive risk management strategies to reduce the risk is advised. Organizations should consider ways to modify or eliminate unacceptable risks.</u>
<u>2 - High</u>	<u>Activities in this category contain unacceptable levels of risk, including catastrophic and critical injuries that are highly likely to occur. Organizations should consider whether they should eliminate or modify activities that still have a "high" rating after applying all reasonable risk management strategies.</u>
<u>1 - Extreme</u>	<u>Activities in this category should not be allowed to proceed without very careful planning. The company needs to evaluate whether the activity is actually necessary in the first place.</u>

Step 3: Develop Safe Practices

Once the risk has been assessed, the appropriate controls need to be put into place.

The main ways to control a hazard include:

- Elimination (including substitution): remove the hazard from the workplace.
- Engineering Controls: includes designs or modifications to plants, equipment, ventilation systems, and processes that reduce the source of exposure.

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- **Administrative Controls:** controls that alter the way the work is done, including timing of work, policies and other rules, and work practices such as standards and operating procedures (including training, housekeeping, and equipment maintenance, and personal hygiene practices).
- **Personal Protective Equipment:** equipment worn by individuals to reduce exposure such as contact with chemicals or exposure to noise.

These methods are also known as the "hierarchy of control" because they should be considered in the order presented (it is always best to try to eliminate the hazard first, etc.).

Controls are usually placed:

1. At the source (where the hazard "comes from")
2. Along the path (where the hazard "travels")
3. At the worker

Control at the source and control along the path are also known as engineering controls.

Administrative controls limit workers' exposure by implementing other "rules", such as training, supervision, shorter shifts in high risk areas etc. These control measures have many limitations because the hazard itself is not actually removed or reduced. Administrative controls are not generally favored because they can be difficult to implement, maintain and are not a reliable way to reduce exposure

Personal protective equipment (PPE) includes items such as respirators, protective clothing such as gloves, face shields, eye protection, and footwear that serve to provide a barrier between the wearer and the chemical or material. It is the final item on the list for a very good reason. Personal protective equipment should never be the only method used to reduce exposure except under very specific circumstances because PPE may "fail" (stop protecting the worker) with little or no warning. For example: "breakthrough" can occur with gloves, clothing, and respirator cartridges.

Once it has been decided what the best and most practical control for a particular hazard is, this needs to be documented. The safe work procedure for the job needs to be written based on those risks and controls

Step 4: Communicate the Controls and Train the Workers

Once the control has been put into place, the workers need to be trained in how to use it. This applies whether it is an engineering control such as a guard or interlock, an administrative control such as a safe work procedure for cold weather or particular PPE when handling a chemical. Training records and/or documented sign-offs are required to show that the workers have been made aware of the hazards and the controls.

Step 5: Review Periodically

Repeat the Hazard Assessment process every 2 years or when site conditions change, when new tasks are added or when new workers join the crew, in order to prevent the development of unsafe working condition.